

Math 4400, Fall 2016 Quiz 2 Name: _____

Calculators, cell phones, or notes are not allowed. Each problem is worth 5 pts.

- (1) Let G be a group. Show that the identity of G is unique and inverses in G are unique.

If e, f are identities in G then $e = e \cdot f = f$.

If $a \in G$ has inverses $b, c \in G$, then $b = b \cdot e = b \cdot a \cdot c = e \cdot c = c$.

- (2) Find 25^{-1} in $(\mathbb{Z}/113\mathbb{Z})^*$ and solve $25x \equiv -4$ modulo 113.

By the Euclidean algorithm

$$113 = 25 \cdot 4 + 13, \quad 25 = 13 + 12, \quad 13 = 12 + 1, \text{ so}$$

$$1 = 13 - 12 = 13 - (25 - 13) = 2 \cdot 13 - 25 = 2 \cdot (113 - 25 \cdot 4) - 25 = 2 \cdot 113 - 9 \cdot 25.$$

Thus $-9 \cdot 25 = 1$ modulo 113. (Equivalently $25^{-1} = -9 = 104$ modulo 113.) Finally we have that

$$x \equiv -9 \cdot 25 \cdot x \equiv -9 \cdot -4 = 36$$

is the only solution of $25x \equiv -4$ modulo 113.